Terminal schedule (review only): Synthetic sample io_marshalling

Revision 02; snapshot schema pp-design/2

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Sources: [user] entered by a user; [data] from the server's design or the configuration's data; [assumption ...] a value its input stated as an assumption, with its statement; [unresolved] the model holds no value.

Board	Owner	Terminal	Kind	Required	Net signal [origin]	Connected to	Template; pattern	Findings
CP-1	A1 (component)	DI.0	signal_di [data]	not required [data]	DI.0 [derived]	X2:1	tenant/synth-io-4di/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	A1 (component)	DI.1	signal_di [data]	not required [data]	DI.1 [derived]	X2:2	tenant/synth-io-4di/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	A1 (component)	DI.2	signal_di [data]	not required [data]	DI.2 [derived]	X2:3	tenant/synth-io-4di/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	A1 (component)	DI.3	signal_di [data]	not required [data]	— [no net]	—	tenant/synth-io-4di/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	A1 (component)	L+	supply_plus [data]	required [data]	24V+ [derived]	X2:1, X2:2, X2:3, X2:4	tenant/synth-io-4di/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	A1 (component)	M	supply_minus [data]	required [data]	— [no net]	—	tenant/synth-io-4di/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed, unconnected_required_terminal
CP-1	X2 (strip)	1 inner p1 L1	signal_di [data]	— [n/a: strip point]	DI.0 [derived]	A1:DI.0	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	1 outer p1 L1	signal_di [data]	— [n/a: strip point]	— [no net]	—	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	1 inner p1 L2	supply_plus [data]	— [n/a: strip point]	24V+ [derived]	A1:L+	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	1 outer p1 L2	supply_plus [data]	— [n/a: strip point]	— [no net]	—	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	2 inner p2 L1	signal_di [data]	— [n/a: strip point]	DI.1 [derived]	A1:DI.1	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	2 outer p2 L1	signal_di [data]	— [n/a: strip point]	— [no net]	—	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	2 inner p2 L2	supply_plus [data]	— [n/a: strip point]	24V+ [derived]	A1:L+	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	2 outer p2 L2	supply_plus [data]	— [n/a: strip point]	— [no net]	—	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed

Board	Owner	Terminal	Kind	Required	Net signal [origin]	Connected to	Template; pattern	Findings
CP-1	X2 (strip)	3 inner p3 L1	signal_di [data]	— [n/a: strip point]	DI.2 [derived]	A1:DI.2	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	3 outer p3 L1	signal_di [data]	— [n/a: strip point]	— [no net]	—	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	3 inner p3 L2	supply_plus [data]	— [n/a: strip point]	24V+ [derived]	A1:L+	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	3 outer p3 L2	supply_plus [data]	— [n/a: strip point]	— [no net]	—	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	4 inner p4 L1	signal_di [data]	— [n/a: strip point]	— [no net]	—	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	4 outer p4 L1	signal_di [data]	— [n/a: strip point]	— [no net]	—	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	4 inner p4 L2	supply_plus [data]	— [n/a: strip point]	24V+ [derived]	A1:L+	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed
CP-1	X2 (strip)	4 outer p4 L2	supply_plus [data]	— [n/a: strip point]	— [no net]	—	tenant/synth-strip-2level/v1; tenant/synth-di-marshal-4ch/v1	template_unreviewed

Structural findings (35: 1 blocking, 34 warning, 21 unresolved)

Severity	Code	Message	Location	Approval	Execution	Confirm
WARNING	pattern_unreviewed	uses pattern tenant/synth-di-marshal-4ch/v1, which no qualified engineer has reviewed; it may not be executed	pattern=tenant/synth-di-marshal-4ch/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template global/legacy-residual_current_…, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=c-86e37505968f32d1bc40b3d6, template=global/legacy-residual_current_device/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=c-8ace24be0b4dcadee62671d8, template=global/legacy-circuit_breaker/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template global/legacy-fuse/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=c-95c77c284077be5bdc00ebf8, template=global/legacy-fuse/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=c-9b04fa4e8f7f8273b6c90e5a, template=global/legacy-circuit_breaker/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template global/legacy-contactor/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=c-ad7ca8ce121a92392ae60e30, template=global/legacy-contactor/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=c-cc4bfcf7b8fbd9993c04a97e, template=global/legacy-circuit_breaker/v1	does not block approval	execution blocker	yes

Severity	Code	Message	Location	Approval	Execution	Confirm
WARNING	template_unreviewed	uses template global/legacy-drive/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=c-d986b938f81e5a897e3f81b2, template=global/legacy-drive/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template global/legacy-contactor/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=c-ed82570ddfa77e389e9b1a74, template=global/legacy-contactor/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template global/legacy-circuit_breaker/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=c-f30368e7e41687b525fd5691, template=global/legacy-circuit_breaker/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template global/legacy-overload_relay/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=c-fc510ed6273f531944bff43f, template=global/legacy-overload_relay/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template tenant/synth-io-4di/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, component=k-4b7b52c55b8da510ac34f651, template=tenant/synth-io-4di/v1	does not block approval	execution blocker	yes
WARNING	template_unreviewed	uses template tenant/synth-strip-2level/v1, which no qualified engineer has reviewed; it may not be executed	board_id=CP-1, strip=s-dfa8615038c44a9c90787b80, template=tenant/synth-strip-2level/v1	does not block approval	execution blocker	yes
BLOCKING	unconnected_required_terminal	required terminal M of component k-4b7b52c55b8da510ac34f651 is in no net	board_id=CP-1, component=k-4b7b52c55b8da510ac34f651, terminal=m	blocks approval	—	no
WARNING / UNRESOLVED	wire_attribute_unset	wire w-090a29bf8baceddfa7e04d5d has no colour; no rule sets it yet	attribute=colour, board_id=CP-1, wire=w-090a29bf8baceddfa7e04d5d	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-0a4d3a8edfaf2ac3ff1979e6 has no colour; no rule sets it yet	attribute=colour, board_id=CP-1, wire=w-0a4d3a8edfaf2ac3ff1979e6	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-43f67b605bce0b8e202ab0d6 has no colour; no rule sets it yet	attribute=colour, board_id=CP-1, wire=w-43f67b605bce0b8e202ab0d6	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-45ec8f2485916680d1cec5bf has no colour; no rule sets it yet	attribute=colour, board_id=CP-1, wire=w-45ec8f2485916680d1cec5bf	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-884bd1aee40da6e8f9bf32b6 has no colour; no rule sets it yet	attribute=colour, board_id=CP-1, wire=w-884bd1aee40da6e8f9bf32b6	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-d31eae56b7fa011ddd71dacf has no colour; no rule sets it yet	attribute=colour, board_id=CP-1, wire=w-d31eae56b7fa011ddd71dacf	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-d3afe8e7d94657f9ee29f95d has no colour; no rule sets it yet	attribute=colour, board_id=CP-1, wire=w-d3afe8e7d94657f9ee29f95d	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-090a29bf8baceddfa7e04d5d has no cross_section_mm2; no rule sets it yet	attribute=cross_section_mm2, board_id=CP-1, wire=w-090a29bf8baceddfa7e04d5d	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-0a4d3a8edfaf2ac3ff1979e6 has no cross_section_mm2; no rule sets it yet	attribute=cross_section_mm2, board_id=CP-1, wire=w-0a4d3a8edfaf2ac3ff1979e6	does not block approval	—	yes

Severity	Code	Message	Location	Approval	Execution	Confirm
WARNING / UNRESOLVED	wire_attribute_unset	wire w-43f67b605bce0b8e202ab0d6 has no cross_section_mm2; no rule sets it yet	attribute=cross_section_mm2, board_id=CP-1, wire=w-43f67b605bce0b8e202ab0d6	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-45ec8f2485916680d1cec5bf has no cross_section_mm2; no rule sets it yet	attribute=cross_section_mm2, board_id=CP-1, wire=w-45ec8f2485916680d1cec5bf	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-884bd1aee40da6e8f9bf32b6 has no cross_section_mm2; no rule sets it yet	attribute=cross_section_mm2, board_id=CP-1, wire=w-884bd1aee40da6e8f9bf32b6	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-d31eae56b7fa011ddd71dacf has no cross_section_mm2; no rule sets it yet	attribute=cross_section_mm2, board_id=CP-1, wire=w-d31eae56b7fa011ddd71dacf	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-d3afe8e7d94657f9ee29f95d has no cross_section_mm2; no rule sets it yet	attribute=cross_section_mm2, board_id=CP-1, wire=w-d3afe8e7d94657f9ee29f95d	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-090a29bf8baceddfa7e04d5d has no number; no rule sets it yet	attribute=number, board_id=CP-1, wire=w-090a29bf8baceddfa7e04d5d	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-0a4d3a8edfaf2ac3ff1979e6 has no number; no rule sets it yet	attribute=number, board_id=CP-1, wire=w-0a4d3a8edfaf2ac3ff1979e6	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-43f67b605bce0b8e202ab0d6 has no number; no rule sets it yet	attribute=number, board_id=CP-1, wire=w-43f67b605bce0b8e202ab0d6	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-45ec8f2485916680d1cec5bf has no number; no rule sets it yet	attribute=number, board_id=CP-1, wire=w-45ec8f2485916680d1cec5bf	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-884bd1aee40da6e8f9bf32b6 has no number; no rule sets it yet	attribute=number, board_id=CP-1, wire=w-884bd1aee40da6e8f9bf32b6	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-d31eae56b7fa011ddd71dacf has no number; no rule sets it yet	attribute=number, board_id=CP-1, wire=w-d31eae56b7fa011ddd71dacf	does not block approval	—	yes
WARNING / UNRESOLVED	wire_attribute_unset	wire w-d3afe8e7d94657f9ee29f95d has no number; no rule sets it yet	attribute=number, board_id=CP-1, wire=w-d3afe8e7d94657f9ee29f95d	does not block approval	—	yes

Findings (0 blocking, 11 warning, 8 info)

[WARNING] CP-1 / c2: Mixer: the cable after a drive is protected by the drive's motor overload setting, which is not checked here. (integrity_drive_cable_unchecked; source: policy; engineering confirmation required)

[WARNING] CP-1 / c1: Dosing pump: Ir 8.6 A, the typical Ir of a 4 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate. (motor_current_table; source: policy; engineering confirmation required)

[WARNING] CP-1 / c2: Mixer: Ir 14.15 A from 7.5 kW shaft power at cos phi 0.85, efficiency 0.9; check it against the nameplate. (motor_current_formula; source: policy; engineering confirmation required)

[WARNING] CP-1 / c2: Mixer: the fuses need at least 140 A prospective short-circuit current at the board to operate fast enough. (drive_fuse_fault_level; source: policy; engineering confirmation required)

[WARNING] CP-1: Loads without a power factor were taken at cos phi 0.9. (default_power_factor; source: policy; engineering confirmation required)

[WARNING] CP-1: Contactors are rated at least their breaker's current (AC-1); confirm the utilisation category against the catalogue for motor or lamp loads. (contactor_ac1; source: policy; engineering confirmation required)

[WARNING] CP-1: 3 circuit(s) have no cable length given; their voltage drop is not checked. (voltage_drop_unchecked; source: policy; engineering confirmation required)

[WARNING] CP-1: No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze. (earth_fault_no_ze; source: policy; engineering confirmation required)

[WARNING] CP-1: Each cable's withstand k^2S^2 is in the cable list; compare it with the selected breaker's let-through I^2t at 10 kA from its maker's curve (ABB handbook §2.4). (short_circuit_withstand; source: policy; engineering confirmation required)

[WARNING] CP-1: Circuit rules are this software's defaults, not confirmed by the company's engineers. (rules_unconfirmed; source: policy; engineering confirmation required)

[WARNING] CP-1: Enclosure: none named, since 6 devices have no width (rail_widths_mm in the company settings). (enclosure_unknown_widths; source: policy; engineering confirmation required)

[INFO] CP-1: Group 1 breaker raised to 16 A, 1.6 x the 10 A breaker after it, for discrimination. (discrimination_group_raised; source: policy)

[INFO] CP-1 / c1: Dosing pump: dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2. (starter_table; source: policy)

[INFO] CP-1 / c2: Mixer: ACS880-01-017A-3 for normal duty at 30 °C, behind 40 A aR fuses (ABB ACS880-01 hardware manual (3AUA0000078093)). Use a screened, symmetrical motor cable, earthed 360° at both ends. (drive_selected; source: policy)

[INFO] CP-1: Incomer raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination. (discrimination_incomer_raised; source: policy)

[INFO] CP-1: Breakers are rated for at least 10 kA breaking capacity against 10 kA prospective at the board; each RCCB's conditional short-circuit current with the breaker ahead of it must reach it too. (breaking_capacity; source: policy)

[INFO] CP-1: Leave 1 spare outgoing ways (20 %). (spare_ways; source: policy)

[INFO] CP-1: Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables. (discrimination; source: policy)

[INFO] CP-1: Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation. (cable_conditions; source: policy)